

FENIL PATEL

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Professional Summary

AI/ML Engineer and Big Data Analytics professional with hands-on experience in Computer Vision, Deep Learning, Python, C++, SQL, and cloud platforms. Specializing in real-time surveillance AI, video analytics, and deep learning pipeline development — including line crossing, intrusion detection, ALPR with OCR, and object anomaly detection — with active work on Edge AI deployment, model optimization, and model compression using NPU-based SoCs (Augentix, Kneron, DeepX). Experienced in ISP pipeline integration, embedded systems inference, multi-camera stream processing, and on-device inference optimization for production CCTV and surveillance AI systems. Proven track record building scalable data pipelines and intelligent search engines. Seeking to contribute to Edge AI, Computer Vision, Embedded AI, or Applied ML engineering at a top-tier organization.

Education

Center for Development of Advanced Computing (C-DAC, SMVITA)

February 2025 – July 2025

Post Graduate Diploma in Big Data Analytics

C.K.Pithawala College of Engineering and Technology, Surat

September 2020 – May 2024

Bachelor of Engineering in Computer Engineering

7.32 CGPA (73.2%)

Reliance Foundation School, Surat

March 2020

Higher Secondary Education – Science

81.66 Percentile

Experience

NXON AI PRIVATE LIMITED

September 2024 – Present

AI/ML Developer

Ahmedabad, Gujarat

- Architected and deployed a production-grade AI-powered CCTV event detection engine capable of identifying 15+ real-time security events across simultaneous live multi-camera feeds with sub-100 ms per-frame latency.
- Line Crossing Detection:** Constructed configurable virtual-line crossing detection using optical flow and centroid tracking; supports bidirectional rules with entry and exit counts and intelligent alert suppression logic.
- Region Entry / Exit & Intrusion Detection:** Designed polygon-based region monitoring to flag unauthorized zone entry, exit, and dwell-time violations; integrated real-time alert dispatch via REST API.
- Object Abandonment & Missing Object Detection:** Implemented background-subtraction and object-persistence algorithms to detect abandoned items and flag missing objects from a scene within configurable time thresholds.
- Person Detection:** Fine-tuned YOLO-family models for high-accuracy pedestrian detection under varied lighting and occlusion conditions; achieved 28% mAP improvement over baseline.
- Automatic License Plate Recognition (ALPR) with OCR:** Deployed an end-to-end ALPR pipeline — plate localization, character segmentation, and OCR — achieving reliable plate reads across multiple vehicle types and camera angles in real-world conditions.
- Edge AI / NPU Deployment:** Working with Augentix, Kneron, and DeepX AI SoC platforms; applying INT8 and INT16 quantization, layer fusion, and NPU operator mapping to deploy models fully on-chip, eliminating GPU or cloud dependency.
- NPU Pipeline Optimization:** Leveraged deep knowledge of NPU internal architecture — DMA scheduling, inference engine constraints, memory bandwidth — to restructure model graphs for maximum NPU utilization and fastest AI event triggering on edge hardware.
- Engineered high-performance C++ video-processing modules fused with Python deep-learning inference pipelines; reduced per-frame processing overhead by 35% and deployed REST APIs to stream AI events to dashboards and alerting systems in real time.
- Conducted model training, fine-tuning, and evaluation cycles using PyTorch and TensorFlow; implemented custom deep learning models for pose estimation, motion tracking, and anomaly detection within a unified AI event management system.

NXON AI PRIVATE LIMITED

July 2024 – September 2024

Full-Stack Developer

Ahmedabad, Gujarat

- Employed expertise in DSA, Algorithms, Java, Python, and React.js in solving real-world engineering problems.
- Carried out research on MySQL database optimization, improving query efficiency and reducing data fetching time.
- Maintained and extended web applications using modern frameworks and technologies.
- Diagnosed and resolved 15+ code errors weekly in PHP applications, leveraging debugging tools and real-time monitoring to improve website stability by 40% and reduce user-reported issues.

- Assisted in developing and maintaining web applications using Laravel and PHP.
- Designed and optimized MySQL database schema and queries.
- Supported frontend integration with Blade templates, HTML, CSS, and JavaScript.
- Participated in code reviews and followed version control practices using Git.
- Executed testing, debugging, and documentation of modules. Collaborated with senior developers in delivering scalable and secure solutions.

Projects

AI-Powered CCTV Event Detection System | OpenCV, TensorFlow, PyTorch, C++, C, Python, REST API

- End-to-end surveillance AI platform detecting 15+ event types — line crossing, region intrusion, loitering, abandoned or missing object detection, person detection, and ALPR with OCR — across live multi-camera CCTV feeds.
- Deployed production models on Augentix, Kneron, and DeepX NPU-based edge SoCs using INT8 quantization and layer fusion; eliminated cloud or GPU dependency while achieving sub-100 ms inference latency per frame.
- Built modular C++ video-processing layer coupled with Python ML pipelines; streamed AI events to real-time monitoring dashboards via REST APIs.

Deeplense – AI-Powered Semantic Image Search Engine | Python, CLIP, FastAPI, Next.js, Qdrant, PostgreSQL

- Architected a production-ready image search application combining semantic search (CLIP embeddings), keyword search (BM25), and intelligent filtering to enable natural-language-based image retrieval.
- Hybrid ranking algorithm — semantic matching (60%), BM25 keyword scoring (25%), metadata boosting (15%) — attained 40–60% accuracy improvement over keyword-only search.
- FastAPI backend with Qdrant vector DB storing 512-D CLIP embeddings; 50–100 ms search latency over 5,000+ images with 20–30 images/sec GPU indexing throughput.

Human Decision Simulator – NARS Cognitive Agent | Python, Streamlit, NARS, Layered Architecture

- Advanced cognitive simulation engine using a Non-Axiomatic Reasoning System (NARS) for bounded-rationality multi-goal arbitration; multi-hop deductive inference with action confidence scores exceeding 0.84.
- Clean layered architecture (domain → application → infrastructure → interface) with real-time Streamlit GUI, rich CLI, and comprehensive unit and integration test coverage.

POLISCAN – Election Contributions Analytics | AWS, Python, Pandas, Power BI, Cloud, Jira, GitHub

- Performed in-depth analysis of financial contributions during U.S. elections by integrating data related to individuals, candidates, and committees; uncovered donor behavior trends, party-wise comparisons, and regional donation patterns.
- Created an interactive Power BI dashboard to present key insights on financial flows and behavioral dynamics within the political donation ecosystem.

Technical Skills

Programming Languages: Python, C++, C, TypeScript, JavaScript, Java, R, SQL, PHP, HTML/CSS

Frameworks & Libraries: OpenCV, TensorFlow, PyTorch, CLIP, YOLO, Scikit-learn, FastAPI, Django, Laravel, React.js, Next.js, Streamlit, Bootstrap, NARS

AI / Machine Learning: Computer Vision, Video Analytics, Surveillance AI, Object Detection, Pose Estimation, Optical Flow, ALPR and OCR, Deep Learning, NLP, Model Compression, INT8 and INT16 Quantization, Layer Fusion

Edge AI & Embedded: Augentix, Kneron, DeepX, NPU Operator Mapping, On-Device Inference, Real-Time Inference, ISP Pipeline Integration, Embedded Systems, Edge Deployment, Model Optimization

Cloud, Data & Databases: AWS (S3, EC2, IoT Core), Apache Spark, Pandas, Power BI, Tableau, MySQL, PostgreSQL, MongoDB, Qdrant (Vector DB)

Developer Tools: Git, GitHub, VS Code, PyCharm, Eclipse, Jira, Linux, MySQL Workbench, R Studio

Leadership / Extracurricular

Department of Student Affairs

Student Coordinator

C.K.Pithawala College of Engineering and Technology, Surat, India

- Led campus orientation tours for new students to facilitate their familiarization with university facilities, services, and operations.
- Guided students on how to utilize the services of the university; involved students in university events, sports, and extracurricular activities.
- Delivered a workshop on "How to Work with Arduino".

- Scheduled and inspected all departmental events.
- Conducted a departmental sports festival for approximately 400 students.